Yunzhe Hu

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Education

The University of Hong Kong — Hong Kong, China
Ph.D. in Computer Science (Advisor: Prof. Dong Xu) — *Sept. 2022 – June 2027 (Expected)*

Shanghai Jiao Tong University — Shanghai, China
B.S. in Control Engineering (Artificial Intelligence Track) — *Sept. 2018 – June 2022*

Publications and Preprints (* Equal contribution)

- **Yunzhe Hu**, Dong Xu. VPFlow: Value-Predictive Planning with Imperfect World Models for Offline Goal Reaching. *Under review*, 2026.
- **Yunzhe Hu**, Difan Zou, and Dong Xu. Hyper-SET: Designing Transformers via Hyperspherical Energy Minimization. *ICLR*, 2026.
- **Yunzhe Hu**, Difan Zou, and Dong Xu. An In-depth Investigation of Sparse Rate Reduction in Transformer-like Models. *NeurIPS*, 2024.
- Ming Yang, **Yunzhe Hu**, et al. Complexity-Aware Policy via Heterogeneous Experts for Robotic Manipulation. *IEEE TCSVT*, 2025.
- Guangming Wang*, **Yunzhe Hu***, et al. What Matters for 3D Scene Flow Network. *ECCV*, 2022.
- Guangming Wang*, **Yunzhe Hu***, et al. Residual 3-D Scene Flow Learning With Context-Aware Feature Extraction. *IEEE TIM*, 2022.

Experience

Graduate Research Student — Sept. 2022 – Present
The University of Hong Kong (Advisor: Prof. Dong Xu) — *Hong Kong, China*

- Demonstrated that accurate next-state prediction is unnecessary for effective planning in offline goal reaching, challenging a pervasive assumption; developed *VPFlow* using closed-form hindsight returns to learn value-predictive world models without state reconstruction or bootstrapped targets.
- Identified off-distribution action search as a planning failure mode; designed flow-policy noise steering with imagined advantages to improve goal reaching at test time without weight updates, achieving mean success rate gains of at least 10.1 percentage points on state-based and pixel-based OGBench datasets.
- Formulated *Hyper-SET*, a practical Transformer design framework, from a maximum-likelihood principle that balances semantic alignment and subspace uniformity, revealing a previously unnoticed connection; derived symmetric attention, feedforward layers, and normalization purely through constrained energy minimization, yielding interpretable recurrent Transformers for reasoning, classification, and masked modeling (*ICLR 2026*).
- Identified theoretical pitfalls in constructing attention layers from sparse rate reduction; derived model variants and linked sparse rate reduction to generalization through empirical complexity analysis, yielding improved regularization within the resulting Transformer family (*NeurIPS 2024*).

Undergraduate Research Assistant — Dec. 2020 – June 2022
Shanghai Jiao Tong University (Advisor: Prof. Hesheng Wang) — *Shanghai, China*

- Introduced context-aware feature pyramids and explicit residual refinement to resolve correspondence ambiguities from large motion and repetitive patterns in point clouds, reducing scene flow estimation error by at least 25% (*IEEE TIM 2022*).
- Designed all-to-all flow embedding with bidirectional reliability validation; benchmarked correspondence and refinement design choices, achieving state-of-the-art scene flow estimation accuracy (*ECCV 2022*).

Software Engineering Intern — July 2021 – Aug. 2021
Shanghai Intelligent and Connected Vehicle R&D Center — *Shanghai, China*

- Built a CarSim–Simulink simulation environment and Python vehicle controller to prototype deep reinforcement learning for autonomous parking with OpenAI Gym.

Technical Skills

Languages: Python, C++, Bash
Frameworks: PyTorch, JAX, NumPy, Hugging Face, ROS
Tools: Linux, Git, Docker, Weights & Biases, Jupyter Notebook, GitHub Actions